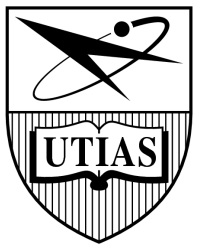
Investigation of Observation Quality on Variational Data Assimilation Methods

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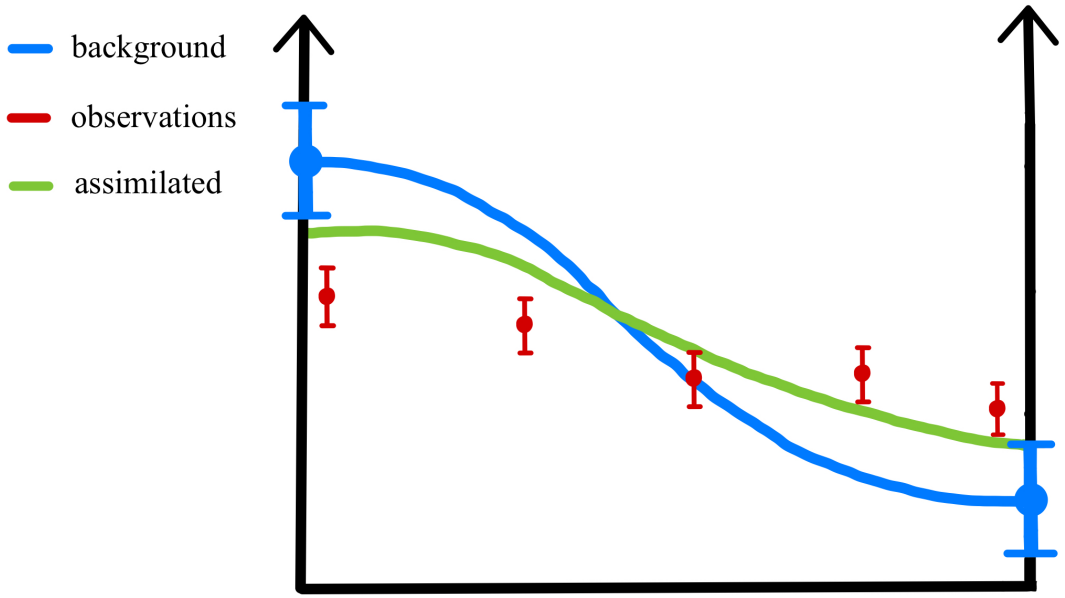




Data Assimilation (DA)

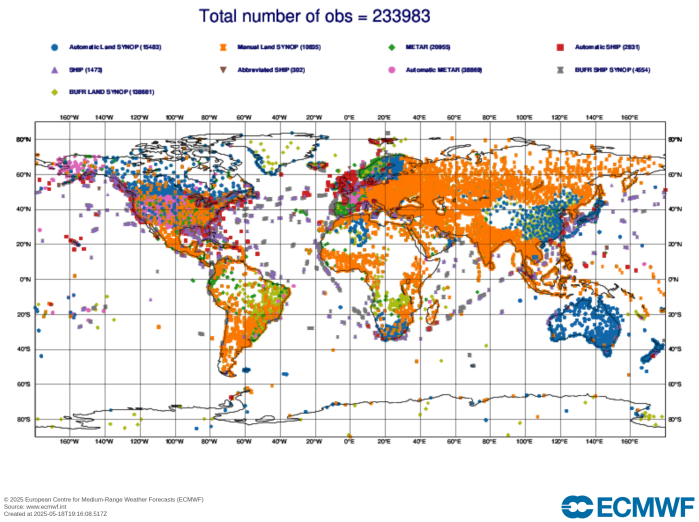
Integrates observed data with numerical models to provide a more accurate simulation of the physical system, and can be used to improve

- **Initial conditions**
- **Boundary conditions**
- Solution parameters
- Model predictions

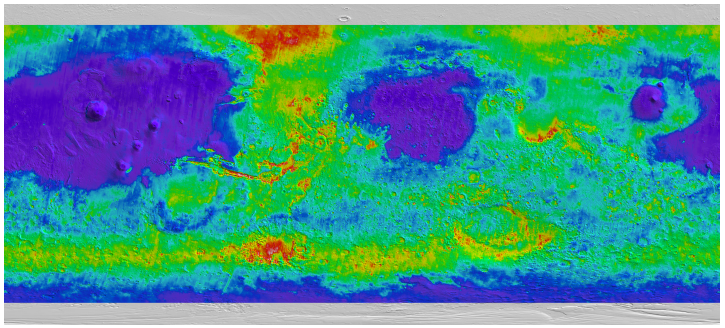


Applications of DA

- Numerical Weather Prediction (NWP)
  - Improve weather forecasts by using atmospheric observations to optimize for the initial conditions of NWP models
  - Example systems: European Centre for Medium-Range Weather Forecasts (ECMWF), Global Forecast System (GFS)
- Mars Global Surveyor (MGS) Mission
  - Understand atmospheric temperature profiles of the Martian atmosphere via collecting thousands of infrared spectra per day



Global observations (ECMWF)



Thermal Emission Spectrometer data (MGS)

DA

Two main classes of DA:

- Sequential (Kalman, 1960)
  - Updates the model around the vicinity of available observations
  - Easier to implement
  - Includes the Kalman filtering approach, a popular implementation of data assimilation
- **Variational** (Le Dimet and Talagrand, 1986)
  - Global fit of assimilation model to the observations, **satisfying model constraints exactly**
  - More successful in data-void regions (Rabier *et al.*, 1992)

Motivation

Purpose/Goal: To provide insight into various observation-related factors affecting the robustness of variational DA algorithms.

Methodology:

- Exploring assimilated solutions of the one-dimensional scalar viscous Burgers' equation
- Steady (**boundary-value**) and unsteady (**initial-value**) problems are studied
- Observation uncertainty, sparsity and frequency (for unsteady problems) are varied to study their effect on the assimilated solution

Variational Data Assimilation

Consider a numerical model, $\mathcal{M}$, as a function of some model parameter, $\mathcal{P}$, describing a state, $\mathbf{U}$, with observations, $\mathbf{d}$

$$\mathbf{U} = \mathcal{M}(\mathcal{P}) + \eta$$
$$\mathcal{P}_{\text{truth}} = \mathcal{P} + \epsilon$$
$$\mathbf{d} = \mathcal{H}(\mathcal{M}(\mathcal{P}_{\text{truth}})) + \xi, \quad \mathcal{H} : \mathbb{R}^{N_{\mathbf{U}}} \rightarrow \mathbb{R}^{N_{\mathbf{d}}}$$

With Gaussian error statistics

$$\epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_{\mathcal{P}})$$
$$\xi \sim \mathcal{N}(0, \mathbf{\Sigma}_{\mathbf{d}})$$

'Strong-constraint' variational approach (Sasaki, 1970): $\eta = 0$

Variational Data Assimilation (cont'd)

Goal: **maximize** the *a posteriori* distribution

$$p(\mathbf{U}|\mathbf{d}) \propto p(\mathbf{d}|\mathbf{U}) \cdot p(\mathbf{U})$$

by the **constrained minimization** of a cost function

$$\begin{aligned} &\underset{\mathcal{P}}{\text{minimize}} \quad J(\mathcal{P}) \\ &\text{subject to } \mathbf{R}(\mathbf{U}) = \mathbf{0} \end{aligned}$$

with

$$\begin{aligned} \textbf{Steady:} \quad \mathbf{R}(\mathbf{U}) &= \frac{\mathrm{d}\mathbf{U}}{\mathrm{d}t} \\ \textbf{Unsteady:} \quad \mathbf{R}^*(\mathbf{U}) &= \mathbf{U}^{n+1} - \mathcal{M}(\mathbf{U}^n) \end{aligned}$$

Variational Data Assimilation (cont'd)

$$\textbf{Steady:} \quad J(\mathcal{P}) = \frac{1}{2}\left\|\mathcal{H}(\mathbf{U}(\mathcal{P})) - \mathbf{d}\right\|^2_{\mathbf{\Sigma}_\mathbf{d}^{-1}} + \frac{1}{2}\left\|(\mathcal{P} - \hat{\mathcal{P}})\right\|^2_{\mathbf{\Sigma}_\mathcal{P}^{-1}}$$

$$\textbf{Unsteady:} \quad J(\mathcal{P}) = \frac{1}{2}\sum_{n=0}^{N}\left\|\mathcal{H}(\mathbf{U}(\mathcal{P})) - \mathbf{d}\right\|^2_{\mathbf{\Sigma}_\mathbf{d}^{-1}} + \frac{1}{2}\left\|(\mathcal{P} - \hat{\mathcal{P}})\right\|^2_{\mathbf{\Sigma}_\mathcal{P}^{-1}}$$

$$= \quad J_{\text{Observation}} \qquad + \quad J_{\text{Background}}$$

with

$$\left\|(\mathcal{P} - \hat{\mathcal{P}})\right\|^2_{\mathbf{\Sigma}_\mathcal{P}^{-1}} \equiv (\mathcal{P} - \hat{\mathcal{P}})^\intercal(\mathbf{\Sigma}_\mathcal{P})^{-1}(\mathcal{P} - \hat{\mathcal{P}})$$

Variational Data Assimilation (cont'd)

Turn into an **unconstrained** problem using Lagrange's method, and introduce **adjoint variables** ($\boldsymbol{\lambda}$)

$$\textbf{Steady: } \mathcal{L} = J + \boldsymbol{\lambda}^{\mathsf{T}} \mathbf{R}$$

$$\textbf{Unsteady: } \mathcal{L} = J + \sum_{n=0}^{N} \left(\boldsymbol{\lambda}^{n+1}\right)^{\mathsf{T}} \mathbf{R}^{*}$$

Setting the derivatives of $\mathcal{L}$ to zero, we are able to retrieve the cost gradient with respect to the parameter, $\mathcal{P}$, of interest

Variational Data Assimilation for IVPs and BVPs

Cost gradient for steady boundary-value problems (BVPs):

$$\frac{\partial J}{\partial \boldsymbol{U}_b} = (\boldsymbol{\Sigma}_{\boldsymbol{U}_b})^{-1}(\boldsymbol{U}_b - \hat{\boldsymbol{U}}_b) - \left[\frac{\partial \boldsymbol{R}}{\partial \boldsymbol{U}_b}\right]^{\mathsf{T}} \boldsymbol{\lambda}$$

and for unsteady initial-value problems (IVPs):

$$\frac{\partial J}{\partial \boldsymbol{U}^0} = (\boldsymbol{\Sigma}_{\boldsymbol{U}^0})^{-1}(\boldsymbol{U}^0 - \hat{\boldsymbol{U}}^0) - \boldsymbol{\lambda}^0$$

Adjoint Variables

To determine the adjoint variables ($\boldsymbol{\lambda}$):

Steady: $$\left[\frac{\partial \boldsymbol{R}}{\partial \mathbf{U}}\right]^{\mathsf{T}} \boldsymbol{\lambda} = \left[\frac{\partial \boldsymbol{\mathcal{H}}}{\partial \mathbf{U}}\right]^{\mathsf{T}} \boldsymbol{\Sigma}_{\mathbf{d}}^{-1} \left(\boldsymbol{\mathcal{H}}(\mathbf{U}) - \mathbf{d}\right)$$

Unsteady: $$\boldsymbol{\lambda}^{n} = \boldsymbol{\lambda}^{n+1} + \Delta t \left[\frac{\partial \boldsymbol{R}(\mathbf{U}^{n+1})}{\partial \mathbf{U}^{n+1}}\right]^{\mathsf{T}} \boldsymbol{\lambda}^{n+1} - \left[\frac{\partial \boldsymbol{\mathcal{H}}^{\boldsymbol{n}}}{\partial \mathbf{U}^{n}}\right]^{\mathsf{T}} \boldsymbol{\Sigma}_{\mathbf{d}}^{-1} \left(\boldsymbol{\mathcal{H}}^{\boldsymbol{n}}(\mathbf{U}^{n}) - \mathbf{d}^{n}\right)$$

optimization: L-BFGS algorithm - ADEPT (ECMWF)
Steady: linear solver: Trilinos
Unsteady: *backward* time-stepping algorithm

One-Dimensional (1D) Scalar Burgers' Equation

1D scalar viscous Burgers' equation, or 1D nonlinear advection-diffusion equation:

$$\frac{\partial u}{\partial t} + \frac{\partial (F - F_v)}{\partial x} = S$$

$$F = \frac{1}{2}u^2,\ F_v = \mu \frac{\partial u}{\partial x},\ S = 0$$

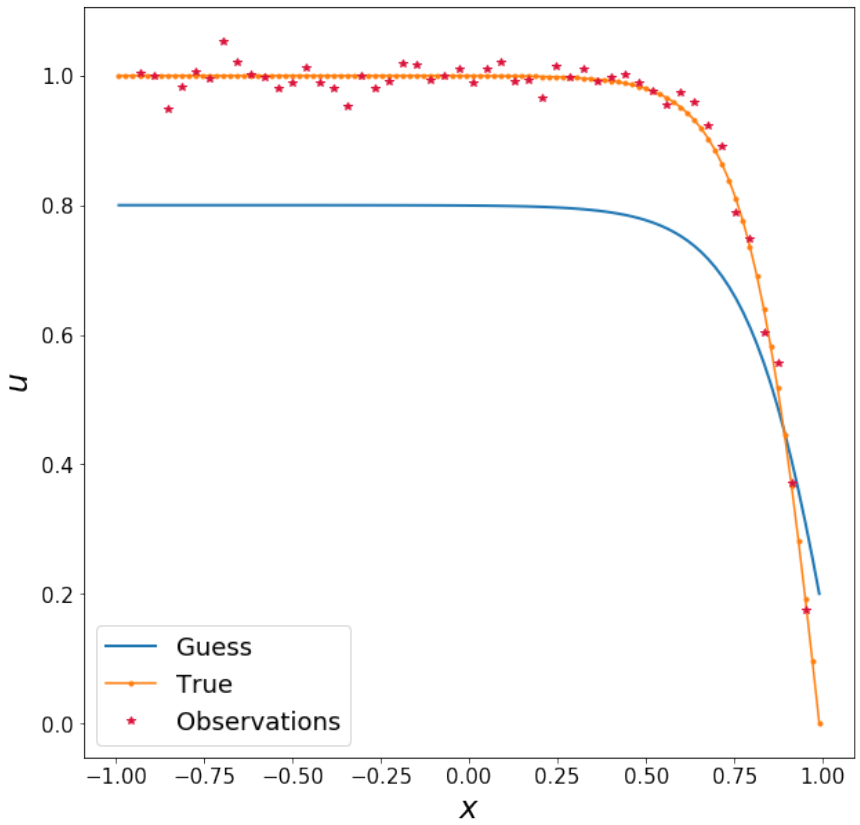
with *true* initial conditions (for **unsteady** problems)

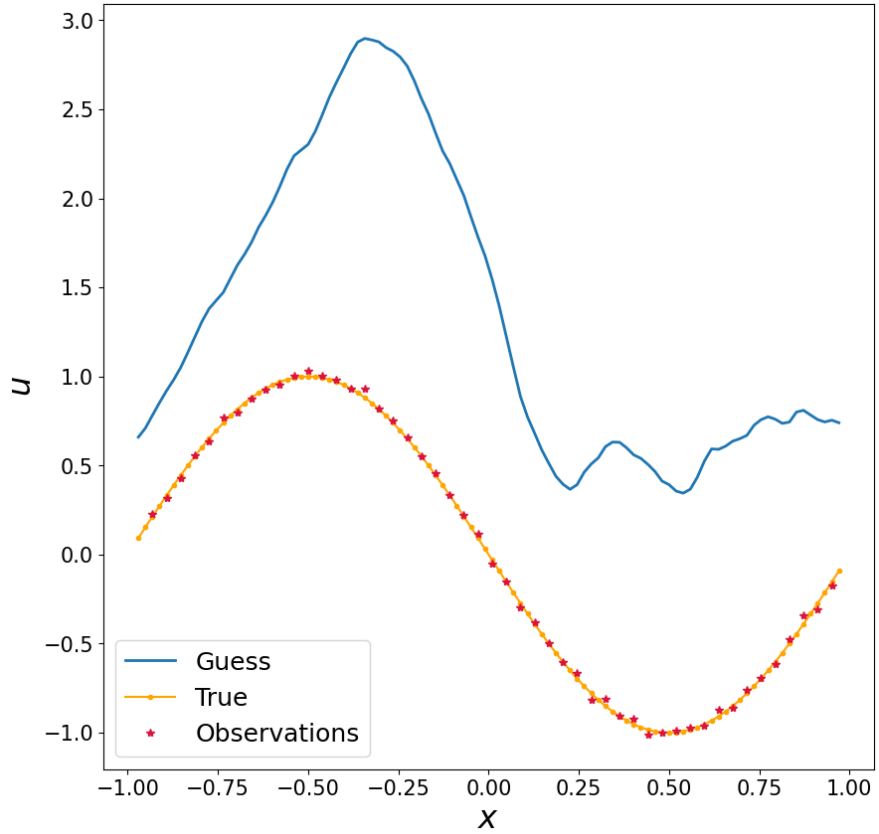
$$u_t(x, 0) = -\sin(\pi x)$$

and *true* boundary conditions (for **steady** problems)

$$u_t(x = 0) = 1, \quad u_t(x = N - 1) = 0$$

Initial and Boundary Conditions



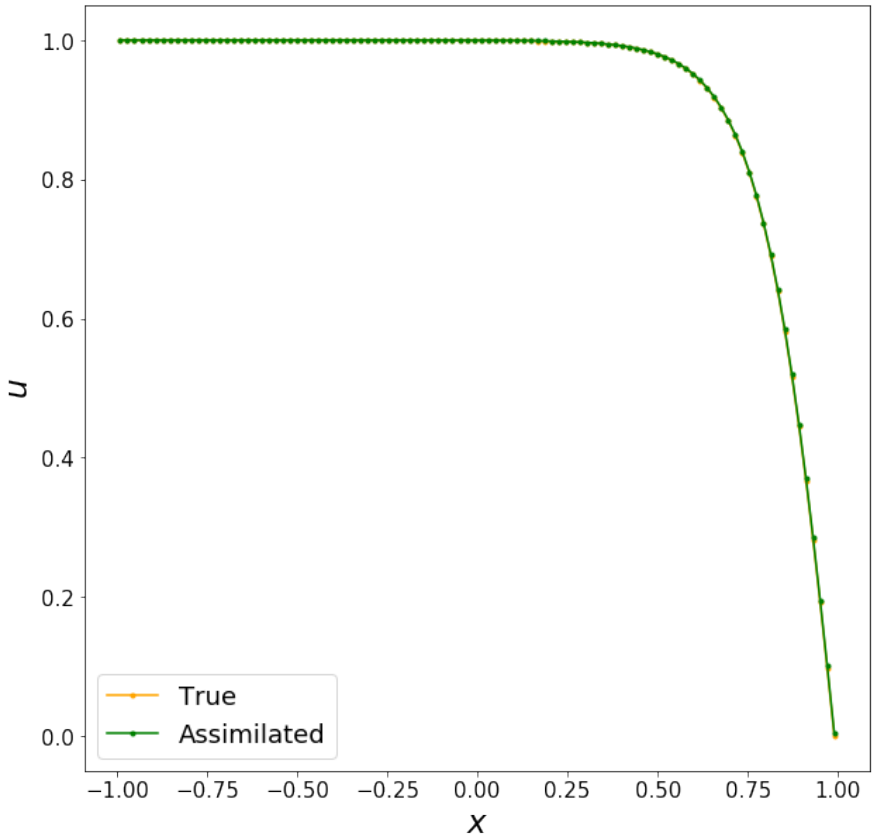


Steady, $\delta_{\text{guess}_L} = -0.2, \delta_{\text{guess}_R} = 0.2$

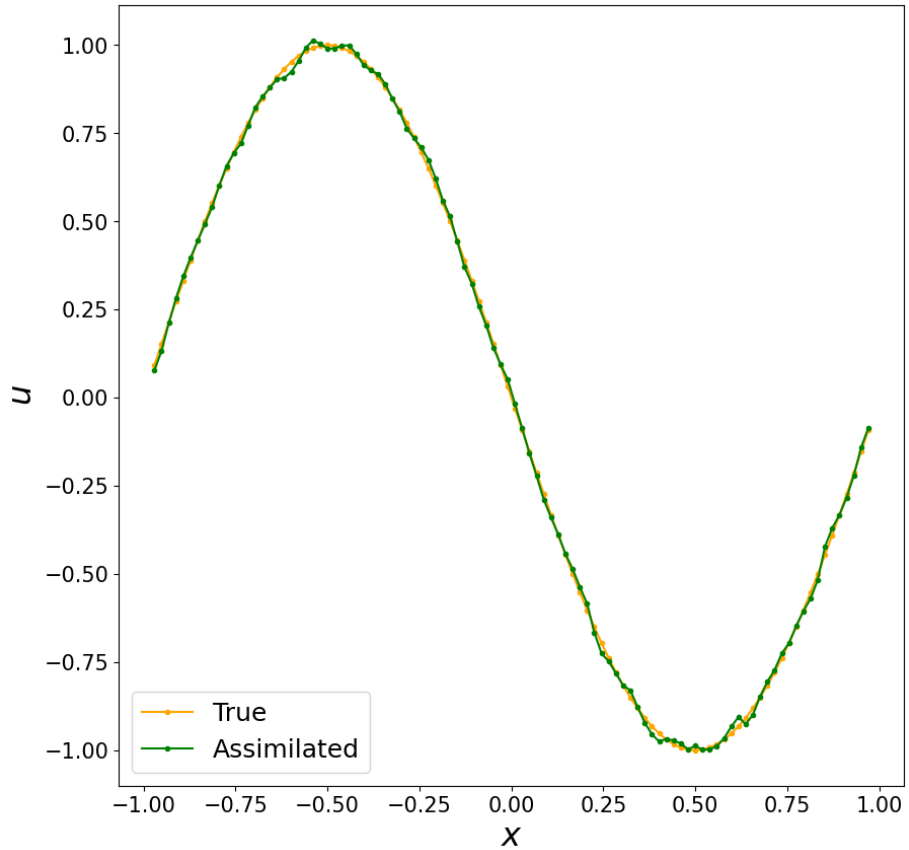
Unsteady, $\vec{\delta}_{\text{guess}} = \vec{\mathbf{b}}$

- observations $= u_{\text{true}} + \mathcal{N}(0, \sigma)$, saved to a file
- sparsity = every 2 cells, standard deviation = 0.02, frequency = 2 time steps

Assimilated Results



Steady

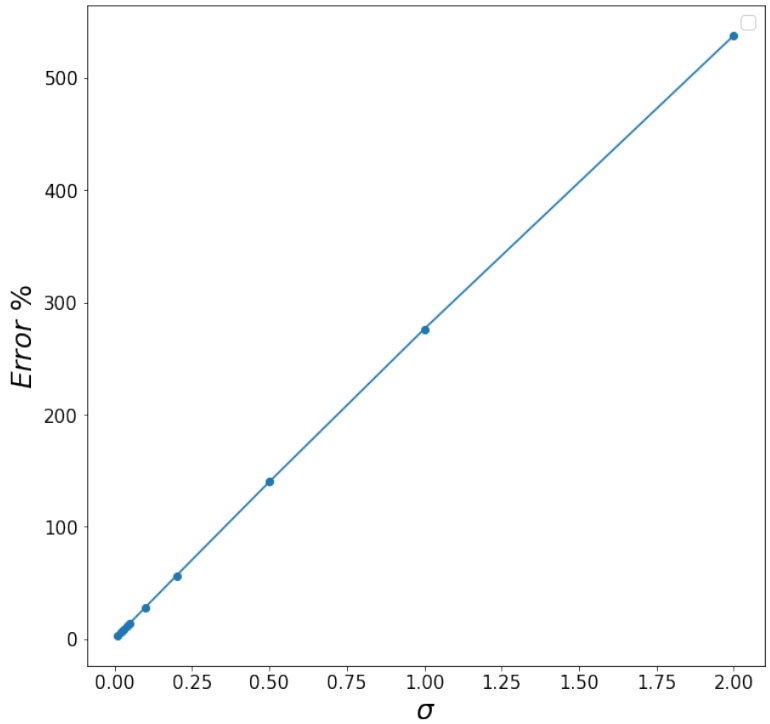


Unsteady

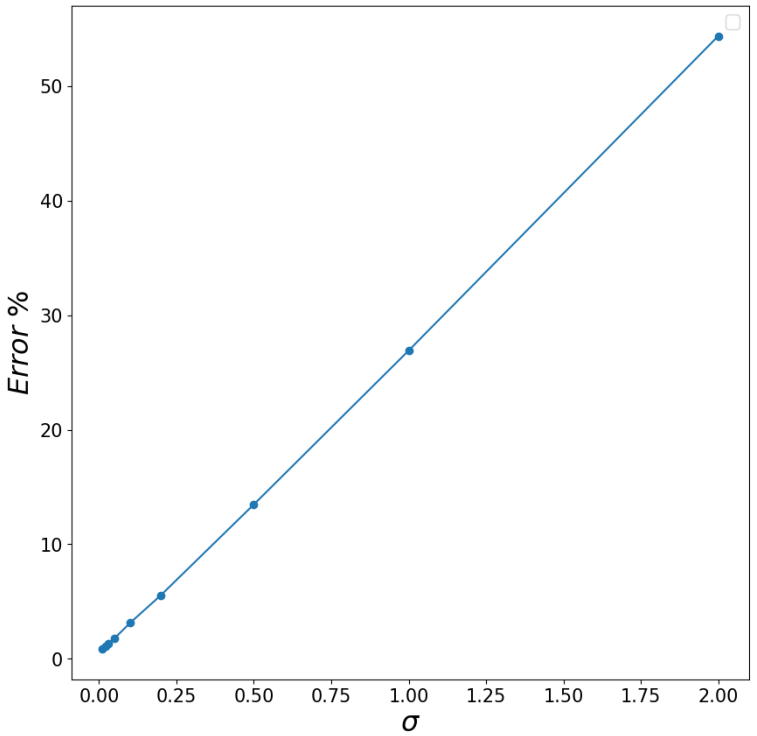
Effect of Observation Uncertainty

We now test how the observation standard deviation affects the *error* of the assimilated solution

$$error = \|u_{\text{true}} - u_{\text{assimilated}}\|_2$$



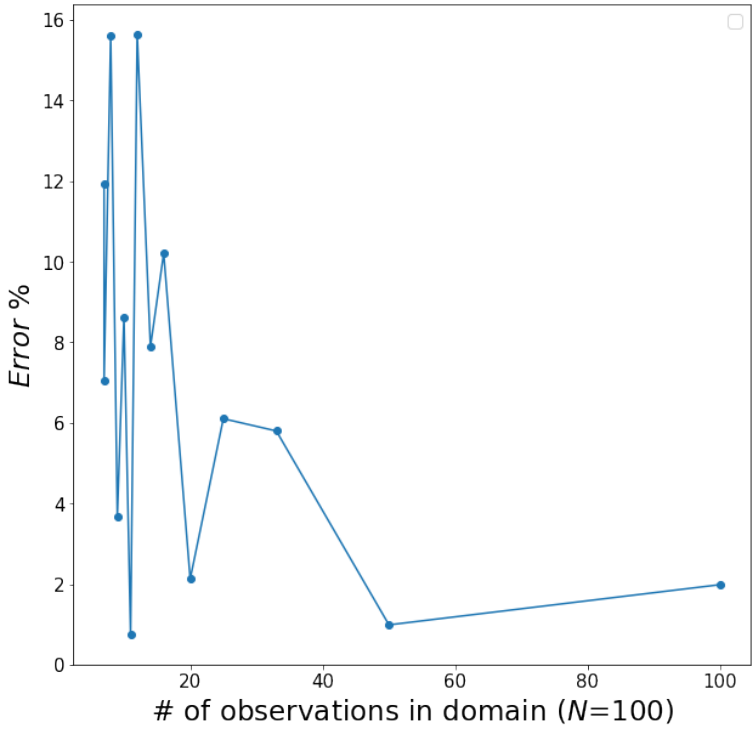
Steady



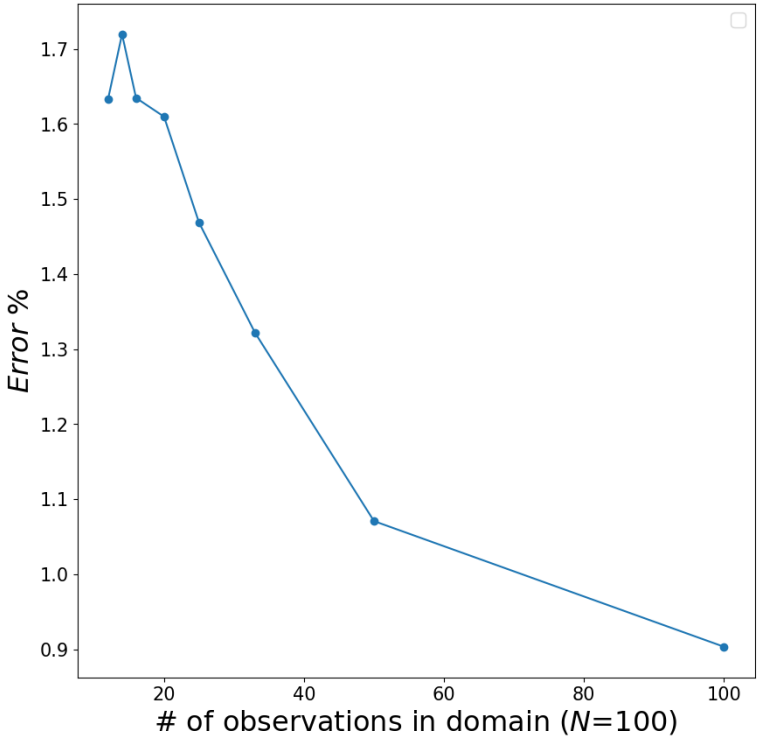
Unsteady

Effect of Observation Sparsity

We now test how the number of observations in the **spatial domain** affects the *error* of the assimilated solution



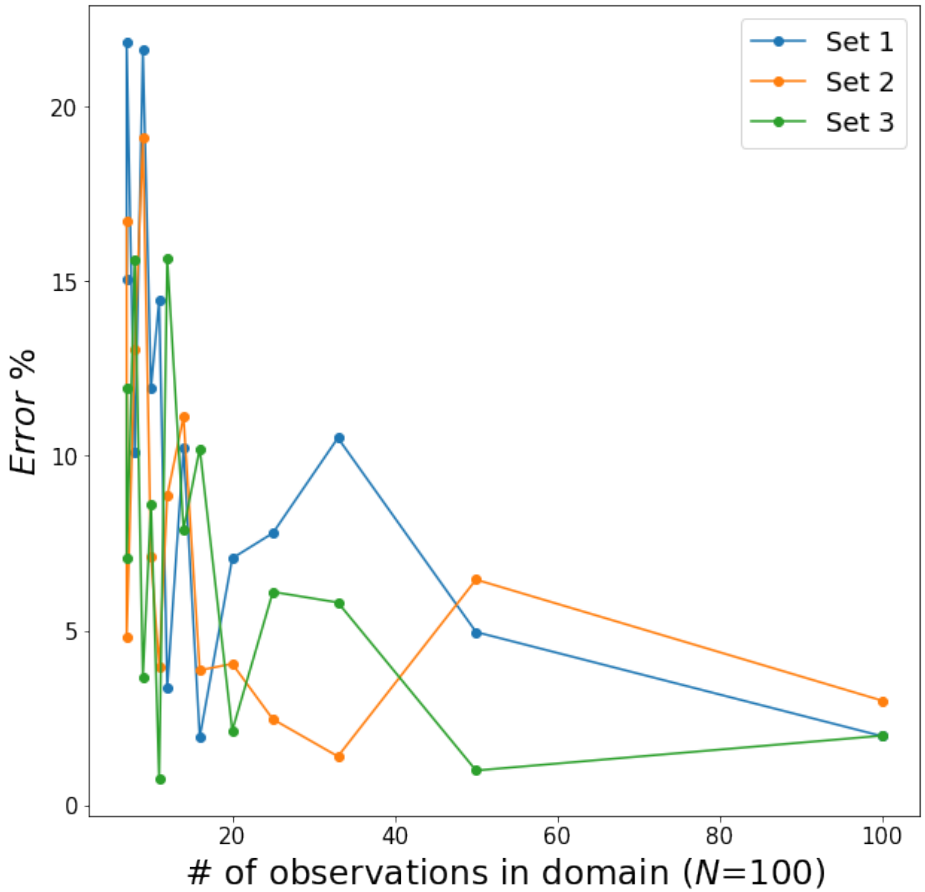
Steady



Unsteady

Effect of Observation Sparsity (cont'd)

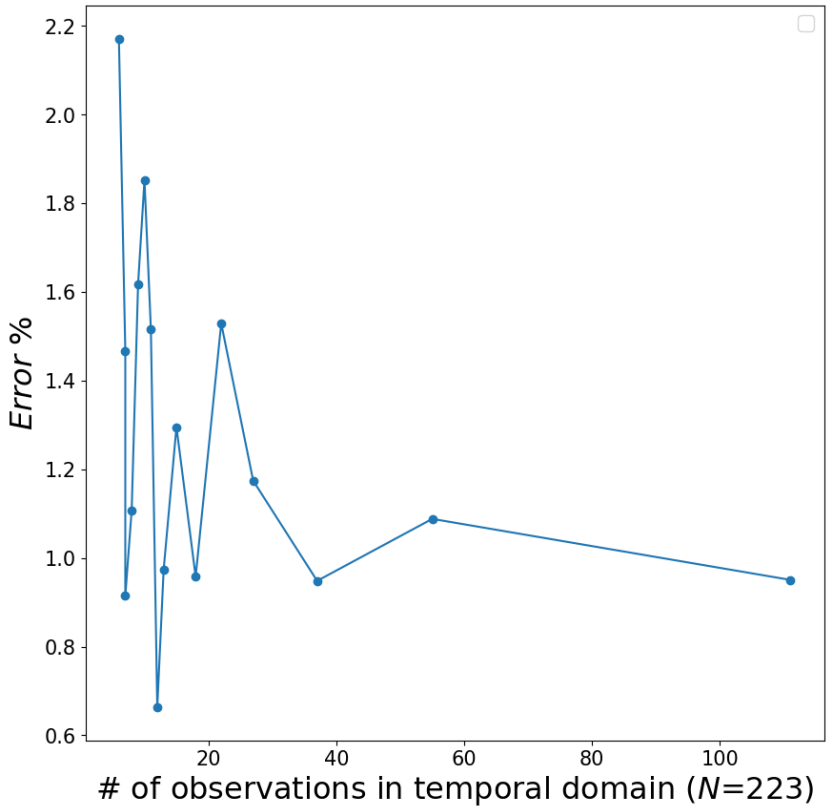
Showcasing the effect of random sampling on observation sparsity



Steady

Effect of Observation Frequency

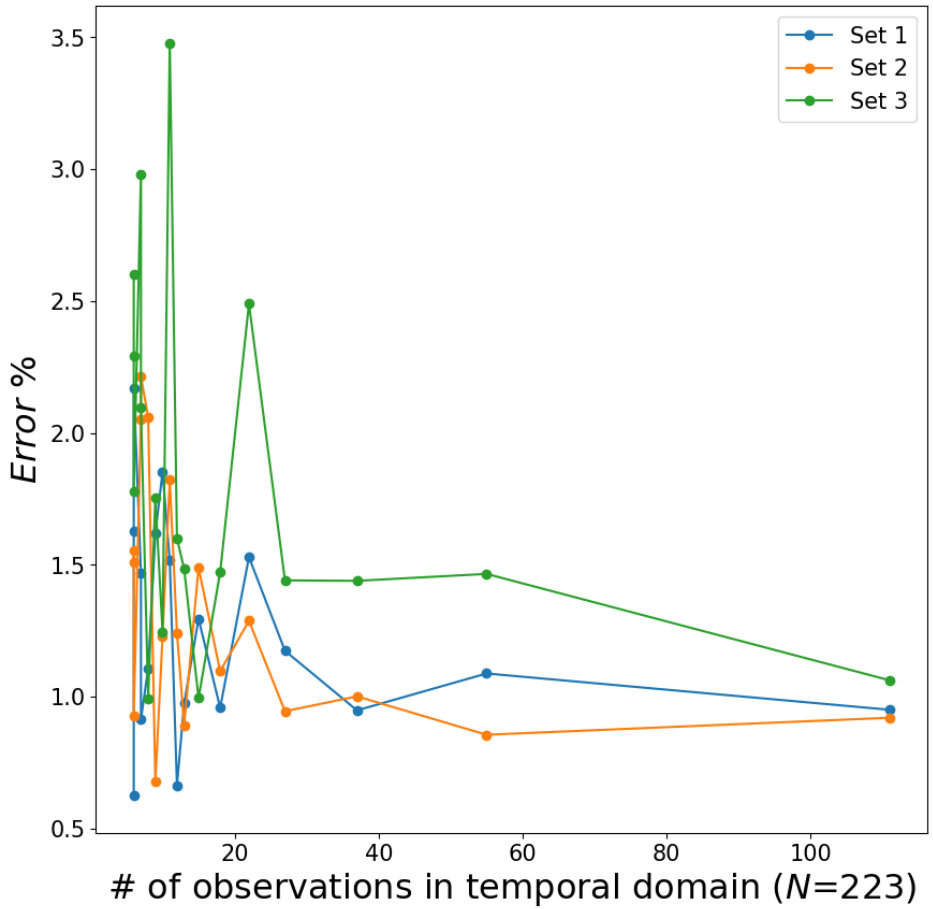
We now test how the number of observations in the **temporal domain** affects the *error* of the assimilated solution (only unsteady case)



Unsteady

Effect of Frequency (cont'd)

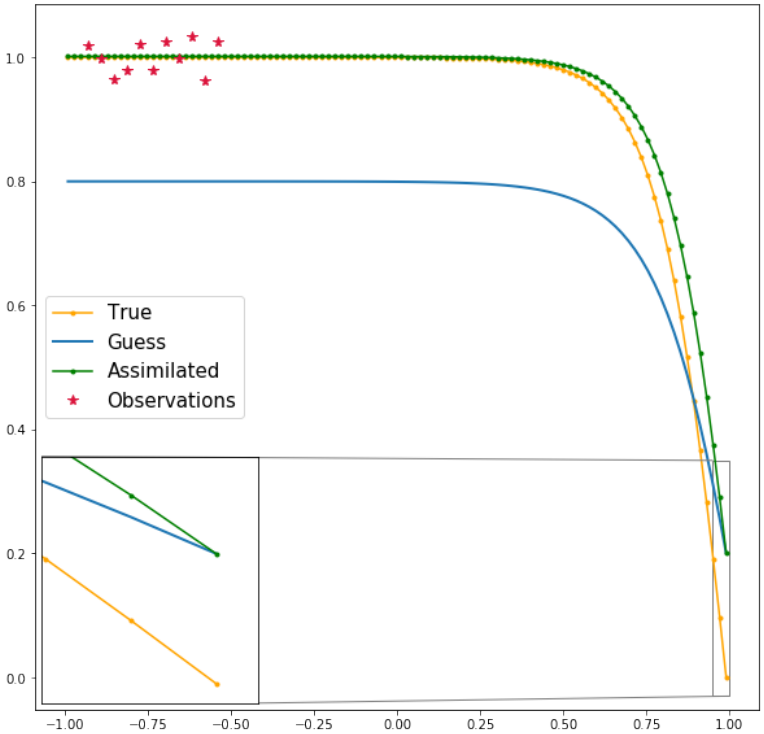
Showcasing the effect of random sampling on observation frequency

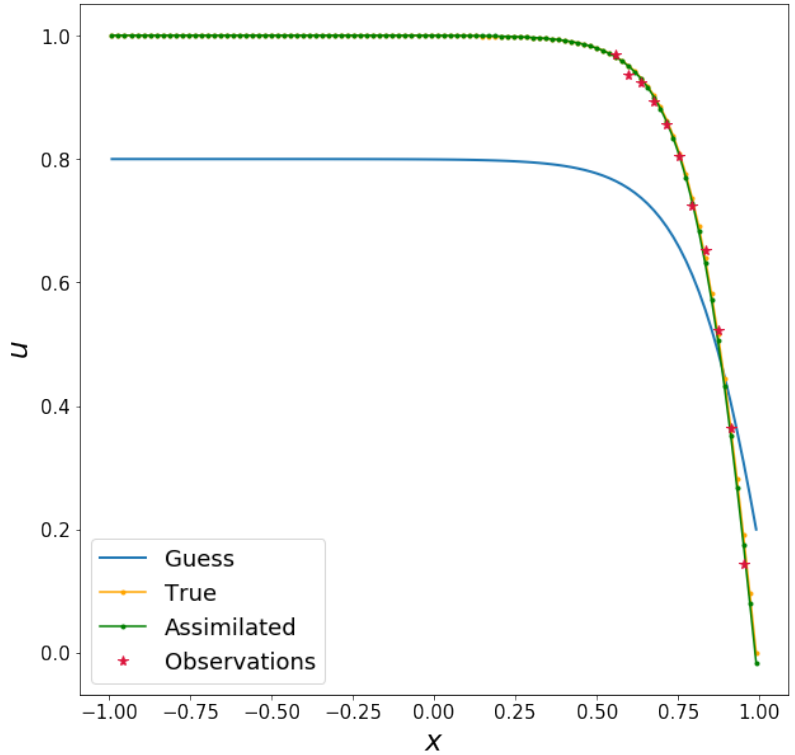


Unsteady

Variational Method Advantage: Sparse Observations

Property of 1D **steady** scalar viscous Burgers' equation: **unique** boundary-values for specific shock location and amplitude/strength





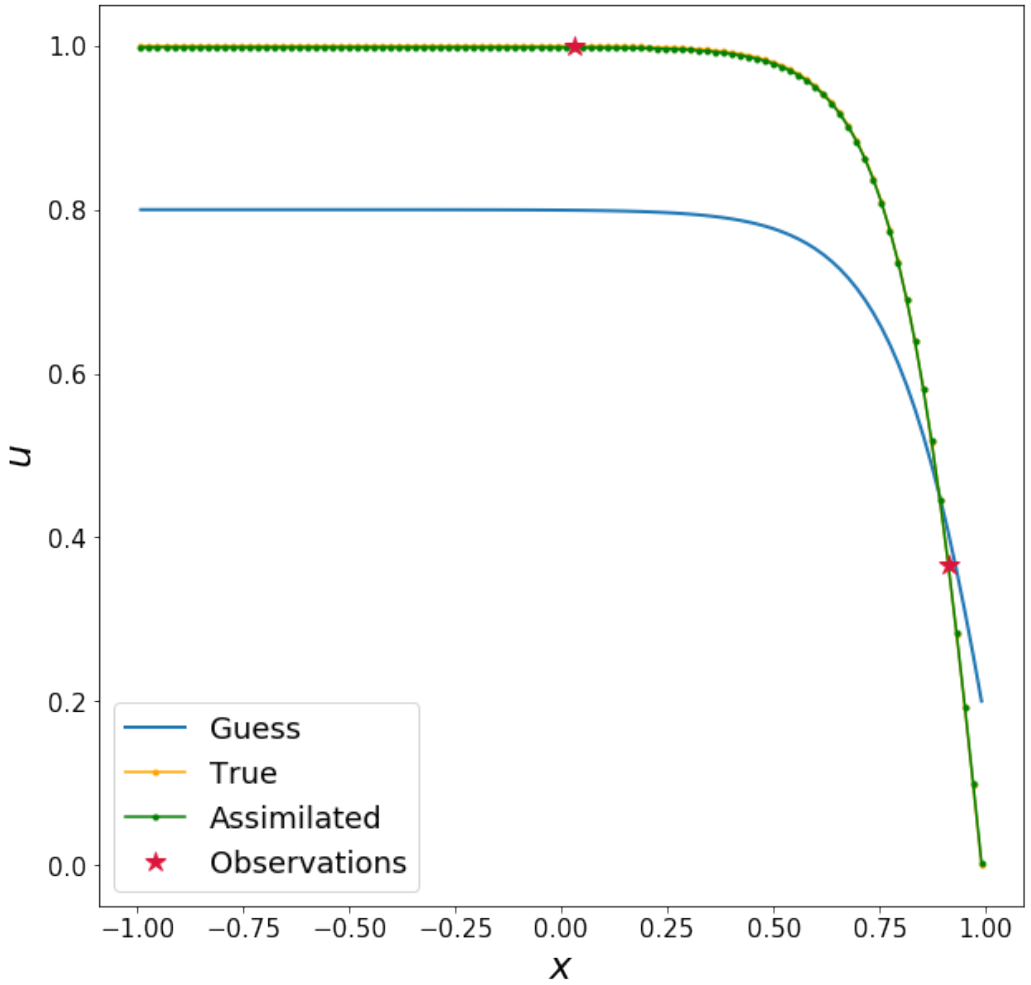
Observations on x <-0.5

Observations on x >0.5

- variational DA respects the **flow physics** (constrained by the model)

Variational Method Advantage: Sparse Observations

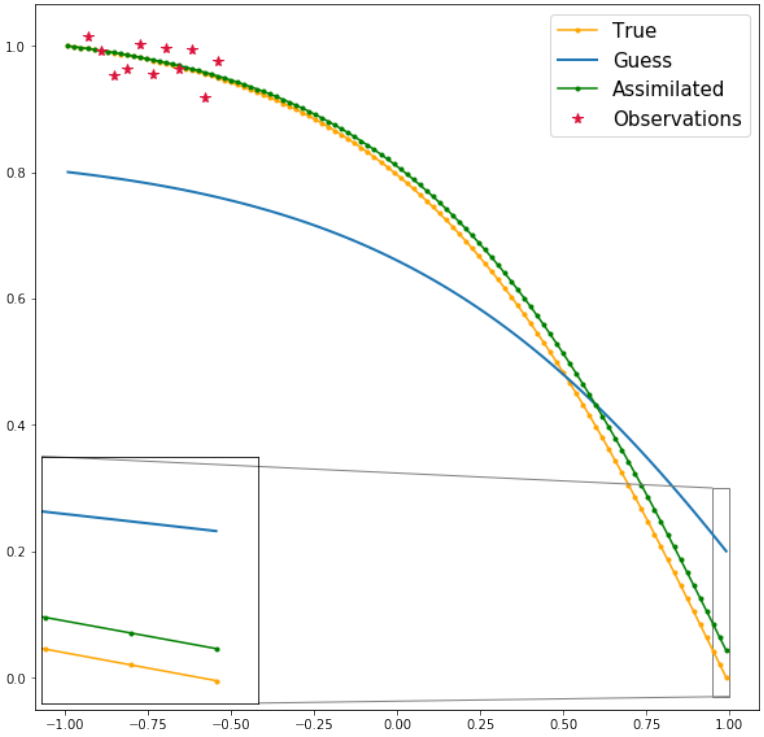
Placing 2 observations with 0 error



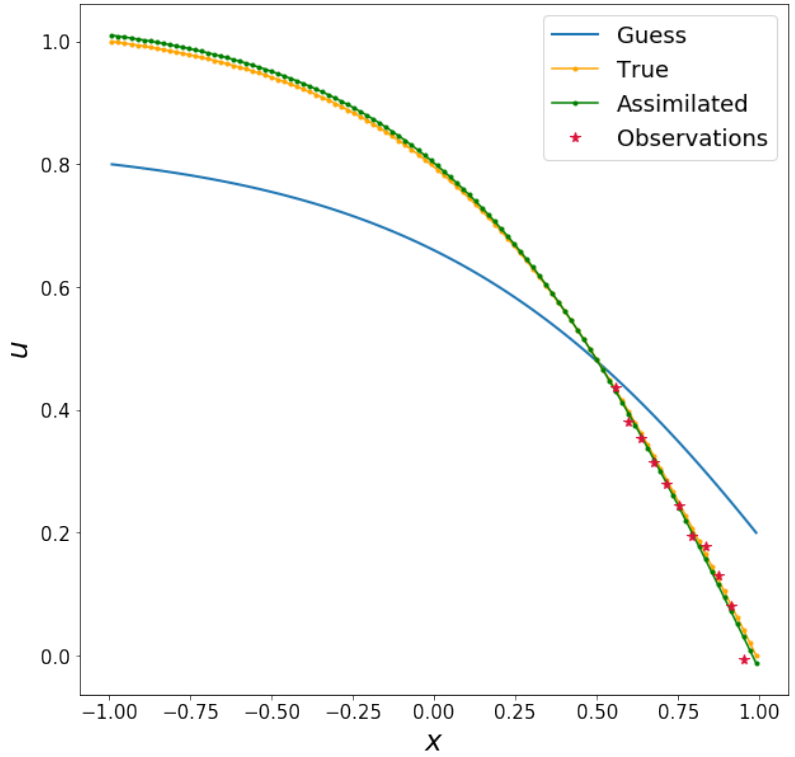
Steady: single observation

Variational Method Advantage: Added Diffusion

More diffusion $\equiv$ solution *spreading out* due to diffusive transport (omni-directional)



Observations on x <-0.5



Observations on x >0.5

- variational DA respects the **propagation of solution information**

Summary and Future Work

We have understood the basic properties of variational-based data assimilation

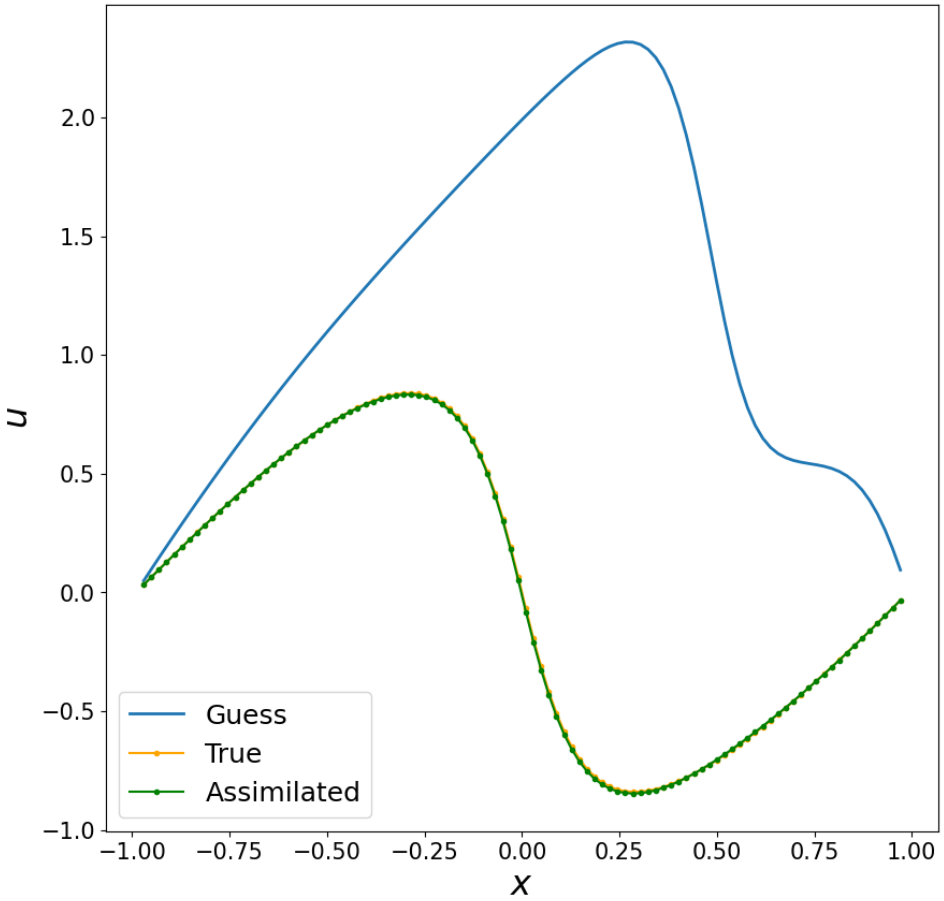
- more observations with a lesser standard deviation is generally better
- assimilated solution performs well in data-void regions, due to being constrained by the model physics
- when more of the solution is transported along the domain, variational methods are able to capture these changes due to them respecting the *flow* of solution information

Future Work:

- Apply variational methods to more complex flows (conservation equations (Navier-Stokes), reactive flows, etc.) with experimental data to improve initial conditions / boundary conditions
- Parameter study with more complex flows (optimizing certain model parameters)

Final Unsteady Assimilated Solution

Final time-step (t=0.3) unsteady assimilated solution



Unsteady at $t = 0.3$

Background Correlation Matrix

A correlation matrix is constructed such that the variances remain unchanged, and covariances approach zero with increasing distance

$$\mathbf{L}(i,j) = \left(1 + \frac{|i-j|\Delta x}{D_L}\right) \exp\left(\frac{-|i-j|\Delta x}{D_L}\right)$$

$D_L \equiv$ localization length scale

Points are then generated via a multivariate normal distribution with mean, $\mu = -\sin(\pi x)$, and variance, $\mathbf{L}$

$N = 100$ samples of $\mathcal{N}(\mu, \mathbf{L})$